1. 设题中矩阵为 A

(1) $A'A=\begin{pmatrix}3&3\\3&3\end{pmatrix}$ ∴取取 $Q=\begin{pmatrix}\frac{1}{\sqrt{2}}&-\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{2}}\end{pmatrix}$ 有 $Q'A'AQ=\begin{pmatrix}6&0\\0&0\end{pmatrix}$

设 $Q=\{\alpha_1,\alpha_2\}$ 取 $\beta_1=\frac{1}{\sqrt{6}}A\alpha_1=(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}})$

补充 $\beta_2=(\frac{1}{\sqrt{2}},0,\frac{1}{\sqrt{2}})$ $\beta_3=(-\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}},\frac{1}{\sqrt{6}})$

∴奇异值分解为 $A=\begin{pmatrix}\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{2}}&-\frac{1}{\sqrt{6}}\\\frac{1}{\sqrt{3}}&0&\frac{2}{\sqrt{6}}\\\frac{1}{\sqrt{3}}&-\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{6}}\end{pmatrix}\begin{pmatrix}\sqrt{6}&0\\0&0\\0&0\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{2}}\\-\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{2}}\end{pmatrix}$

(2) $A'A=\begin{pmatrix}3&2\\2&3\end{pmatrix}$ 取 $Q=\begin{pmatrix}\frac{1}{\sqrt{2}}&-\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}&\frac{1}{\sqrt{2}}\end{pmatrix}$ 则 $Q'A'AQ=\begin{pmatrix}5&0\\0&1\end{pmatrix}$

扫描全能王
3亿人都在用的扫描App

设 $Q=(\alpha_1,\alpha_2)$ ∴ $\beta_1=\frac{1}{\sqrt{10}}A\alpha_1=\left(\frac{2}{\sqrt{10}},\frac{1}{\sqrt{10}},\frac{1}{\sqrt{10}},\frac{2}{\sqrt{10}}\right)'$

$\beta_2=A\alpha_2=(0,\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}},0)$

补充 $\beta_3=(-\frac{1}{\sqrt{2}},0,0,\frac{1}{\sqrt{2}})$ ~~$\beta_4=(0,\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}},0)$~~ $\beta_4=(-\frac{1}{\sqrt{10}},\frac{2}{\sqrt{10}},\frac{2}{\sqrt{10}},-\frac{1}{\sqrt{10}})$

∴ $A=\begin{pmatrix}\frac{2}{\sqrt{10}} & 0 & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{10}}\\ \frac{1}{\sqrt{10}} & \frac{1}{\sqrt{2}} & 0 & \frac{2}{\sqrt{10}}\\ \frac{1}{\sqrt{10}} & -\frac{1}{\sqrt{2}} & 0 & \frac{2}{\sqrt{10}}\\ \frac{2}{\sqrt{10}} & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{10}}\end{pmatrix}\begin{pmatrix}\sqrt{10} & 0\\ 0 & 0\\ 0 & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\end{pmatrix}$

(3) 计算得 $A=\begin{pmatrix}\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{2}}\\ 0 & \frac{2}{\sqrt{6}} & -1 & 0\\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & 0 & 0\\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}\sqrt{6} & 0 & 0\\ 0 & \sqrt{6} & 0\\ 0 & 0 & \sqrt{2}\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}}\\ 0 & 1 & 0\\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}}\end{pmatrix}$

2、(1) $A'A=\begin{pmatrix}2 & 4\\ 4 & 8\end{pmatrix}$

∴ 取 $Q=\begin{pmatrix}\frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}}\\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}}\end{pmatrix}$ ∴ $Q'A'AQ=\begin{pmatrix}10 & 0\\ 0 & 0\end{pmatrix}$

设 $Q=(\alpha_1,\alpha_2)$ ∴ $\beta_1=\frac{1}{\sqrt{10}}A\alpha_1=(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}})'$

取 $\beta_2=(\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}})'$ 仿

∴ $A=\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}\sqrt{10} & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}}\\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}}\end{pmatrix}=\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}}\\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}}\end{pmatrix}\begin{pmatrix}\sqrt{10} & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}}\\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}}\end{pmatrix}$

$=\begin{pmatrix}\frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}}\\ \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}}\end{pmatrix}\begin{pmatrix}\sqrt{10} & 0\\ 0 & 0\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}}\\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}}\end{pmatrix}$

(2) 计算得 $A=\begin{pmatrix}\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2}\\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2}\\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2}\end{pmatrix}\begin{pmatrix}3 & -1 & -1 & -1\\ -1 & 3 & -1 & -1\\ -1 & -1 & 3 & -1\\ -1 & -1 & -1 & 3\end{pmatrix}$

1、$r(A)=r(A'A)\le r(A'A:A'\beta)=r(A'(A:\beta))\le r(A')=r(A)$

∴ $r(A'A)=r(A'A:A'\beta)$ ∴ $A'Ax=A'\beta$ 一定有解